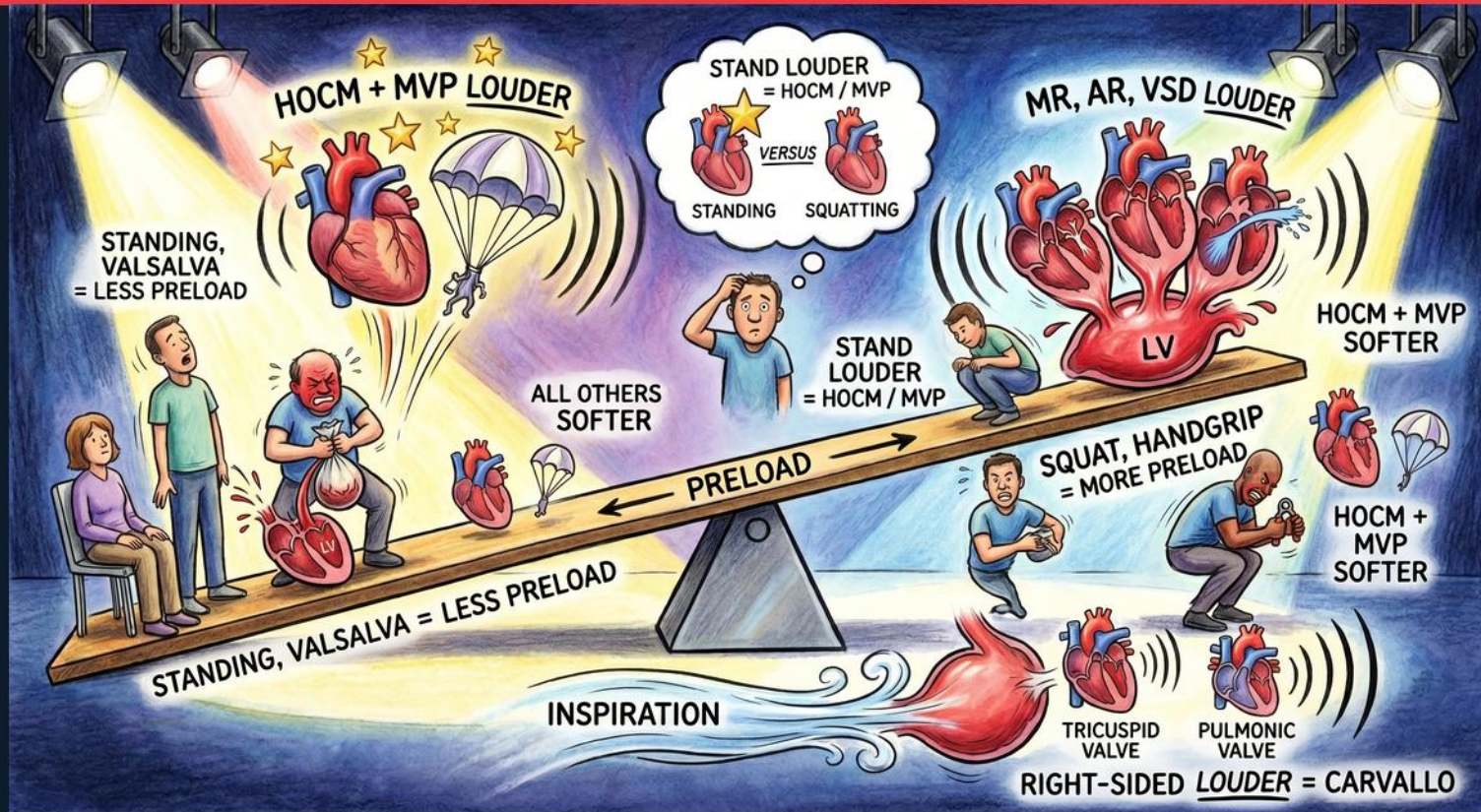
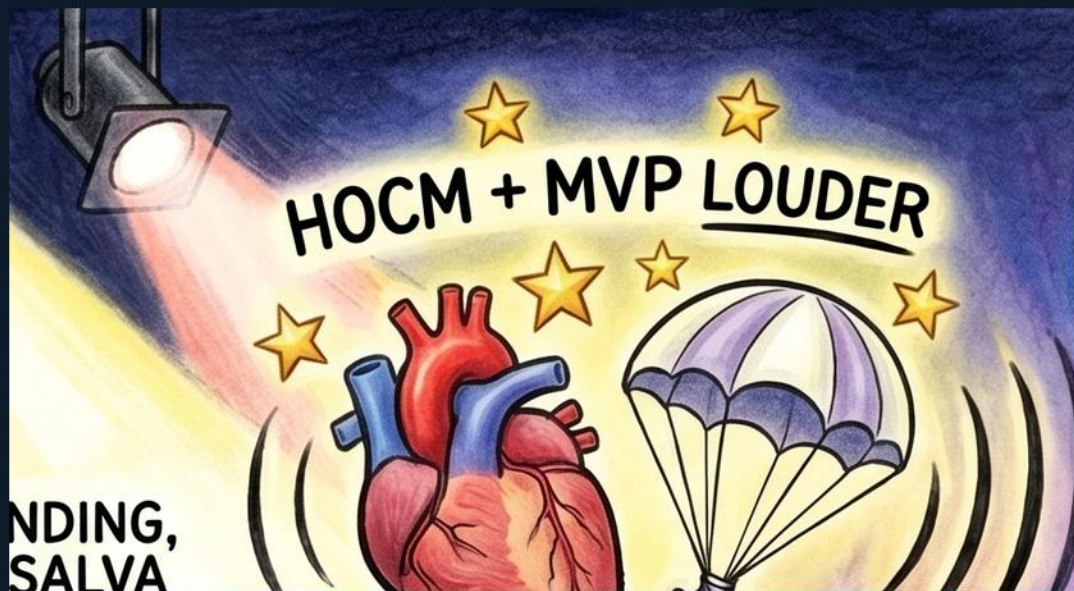


Cardiac Exam — Dynamic Maneuvers





HOCM + MVP LOUDER (top-left banner)

WHAT YOU SEE

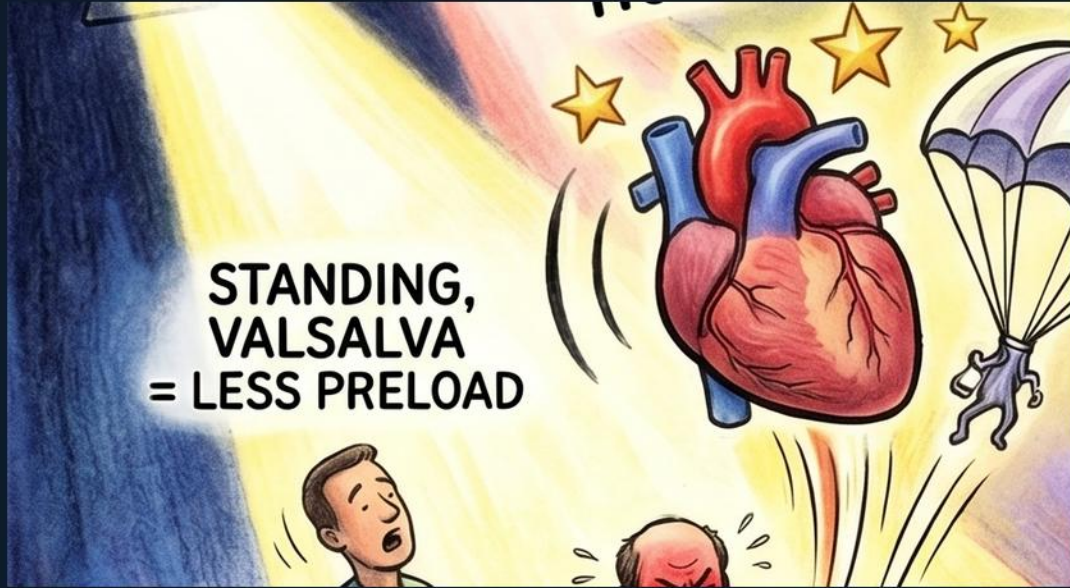
Banner reading HOCM + MVP LOUDER above the standing/Valsalva side of the seesaw.

WHAT IT ENCODES

HOCM and mitral valve prolapse are the EXCEPTION pair: maneuvers that shrink the LV make them LOUDER, the opposite of every other murmur. A smaller left ventricular cavity worsens dynamic LVOT obstruction in HOCM and lets the mitral leaflets prolapse sooner in MVP. Memorize them together as the two that behave backwards.

THE BOARD TRAP

If a murmur gets LOUDER on standing or Valsalva, it is HOCM or MVP — do not call it aortic stenosis or MR, which both soften.



Standing, Valsalva = less preload (figures)

WHAT YOU SEE

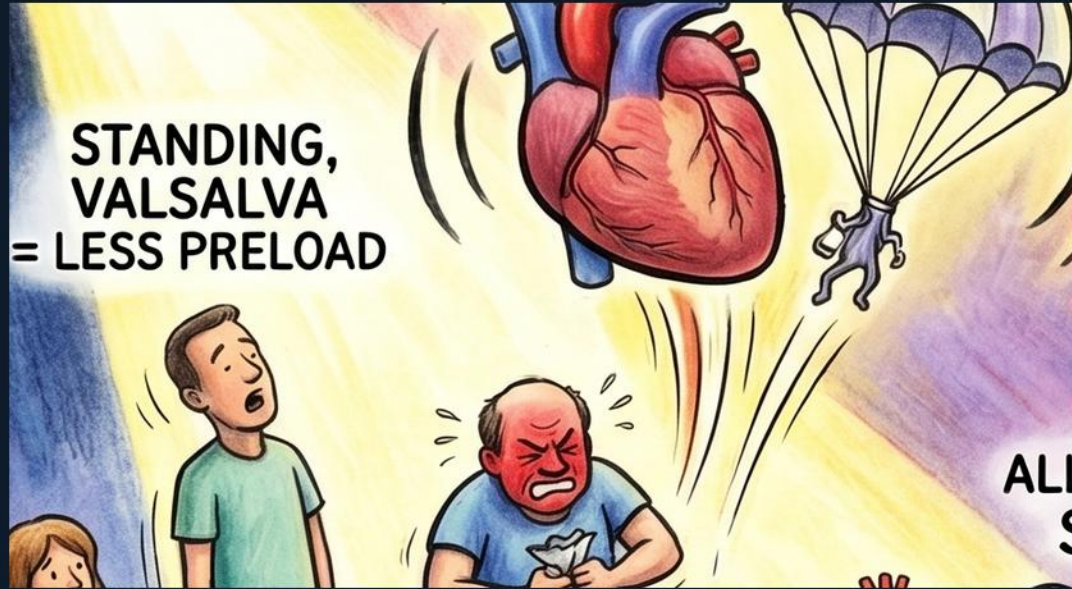
People standing and straining, labeled STANDING, VALSALVA = LESS PRELOAD.

WHAT IT ENCODES

Standing and the strain phase of Valsalva DECREASE venous return and preload, producing a smaller LV. This is the maneuver pair that drives the whole left side of the seesaw. Reduced filling is the unifying mechanism for why most murmurs soften here.

THE BOARD TRAP

Valsalva RELEASE/recovery transiently increases venous return — the classic teaching effect is during the STRAIN phase, not release.



Hypertrophied small-cavity heart (red, angry)

WHAT YOU SEE

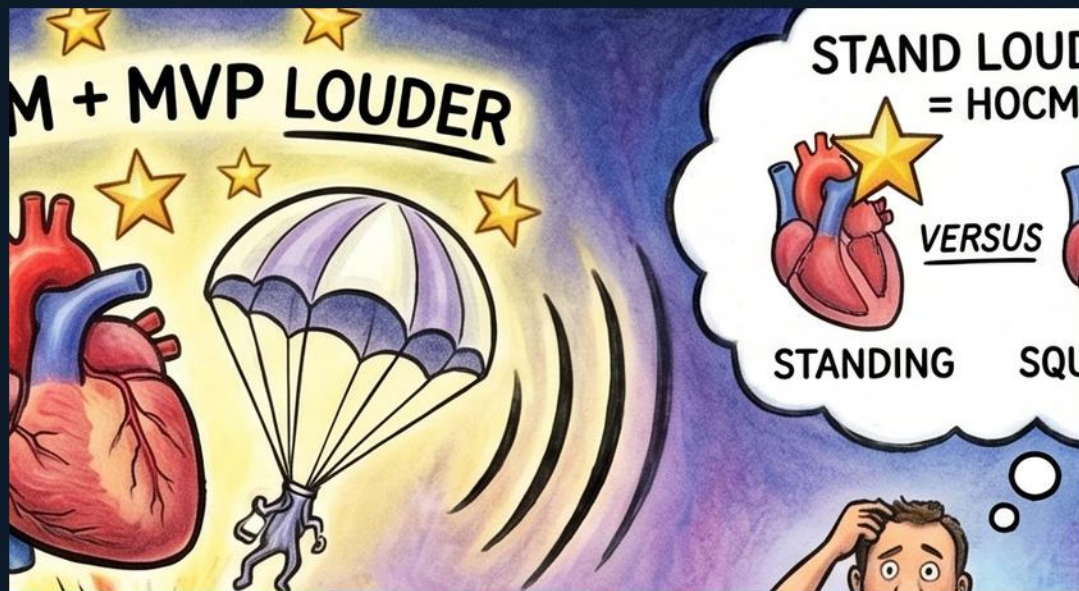
An angry red heart with a thickened wall and a tiny cavity on the standing side.

WHAT IT ENCODES

With less preload the LV cavity collapses smaller. In HOCM the hypertrophied septum and reduced cavity worsen dynamic outflow obstruction, so the systolic murmur grows LOUDER. Smaller chamber size is the through-line for the HOCM/MVP exception.

THE BOARD TRAP

The HOCM murmur intensity tracks degree of obstruction/cavity size, NOT contractility alone — don't assume louder always means worse pump function.



Parachute = mitral valve prolapse

WHAT YOU SEE

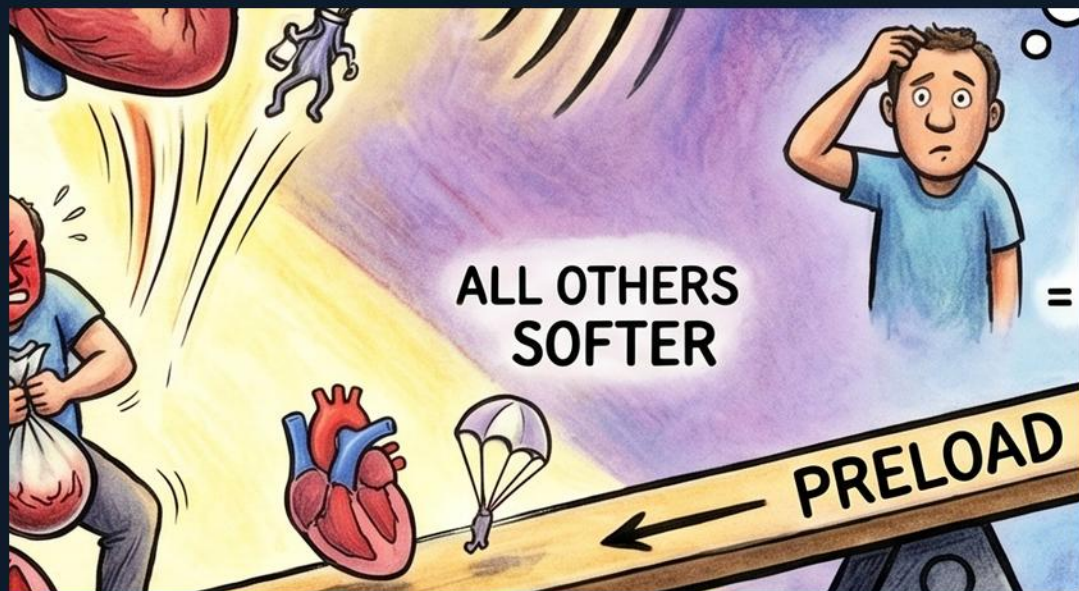
A parachute drawn near the standing side, a visual cue for the billowing prolapsing mitral leaflet.

WHAT IT ENCODES

In MVP, a smaller LV from reduced preload lets the leaflets prolapse EARLIER — the systolic click and murmur move earlier and the murmur lengthens. Standing therefore makes the click-murmur complex come sooner and last longer. MVP joins HOCM as louder-on-standing.

THE BOARD TRAP

Standing moves the MVP click EARLIER (toward S1) and lengthens the murmur — squatting delays the click later; don't reverse the timing.



ALL OTHERS SOFTER

WHAT YOU SEE

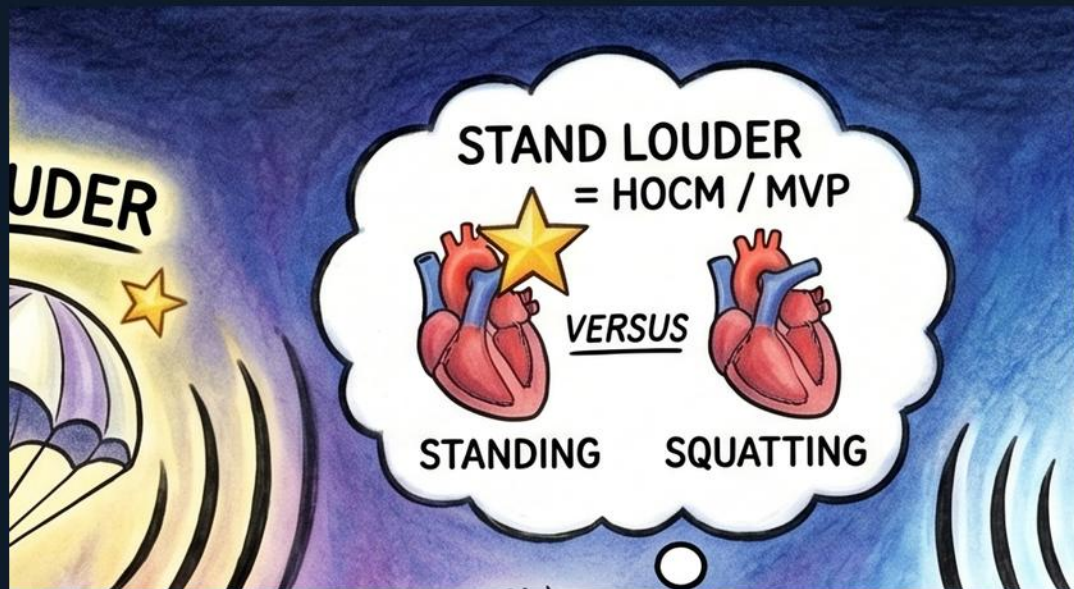
Text ALL OTHERS SOFTER between the two hearts on the standing side.

WHAT IT ENCODES

Every murmur except HOCM and MVP gets SOFTER with standing and Valsalva because reduced preload means less flow across the valve. MR, AR, AS, VSD all diminish. This is the default rule against which HOCM/MVP are the exception.

THE BOARD TRAP

Aortic stenosis softens with Valsalva — a maneuver-induced increase in a systolic murmur points to HOCM, not AS.



Thought bubble: STAND LOUDER = HOCM/MVP

WHAT YOU SEE

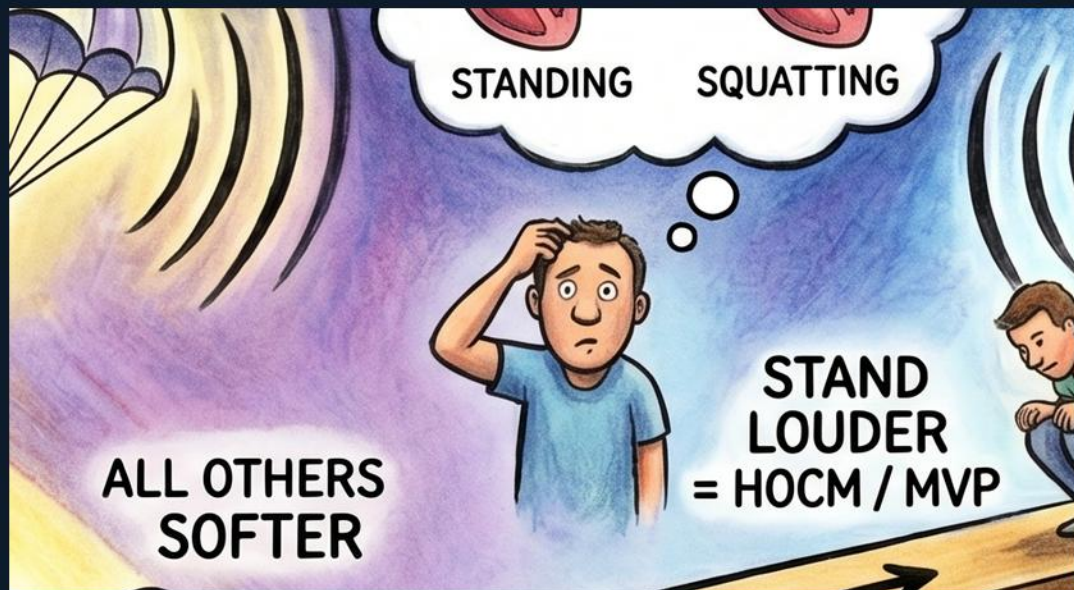
A thinker's thought bubble contrasting STANDING vs SQUATTING with two hearts.

WHAT IT ENCODES

The mnemonic anchor: standing (less preload) = HOCM/MVP louder; squatting (more preload) = HOCM/MVP softer. The two maneuvers are mirror images across the preload axis. Use this single bubble to derive the whole scene.

THE BOARD TRAP

Passive leg raise mimics squatting (raises preload), not standing — both increase venous return and soften HOCM/MVP.



STAND LOUDER = HOCM/MVP (center figure)

WHAT YOU SEE

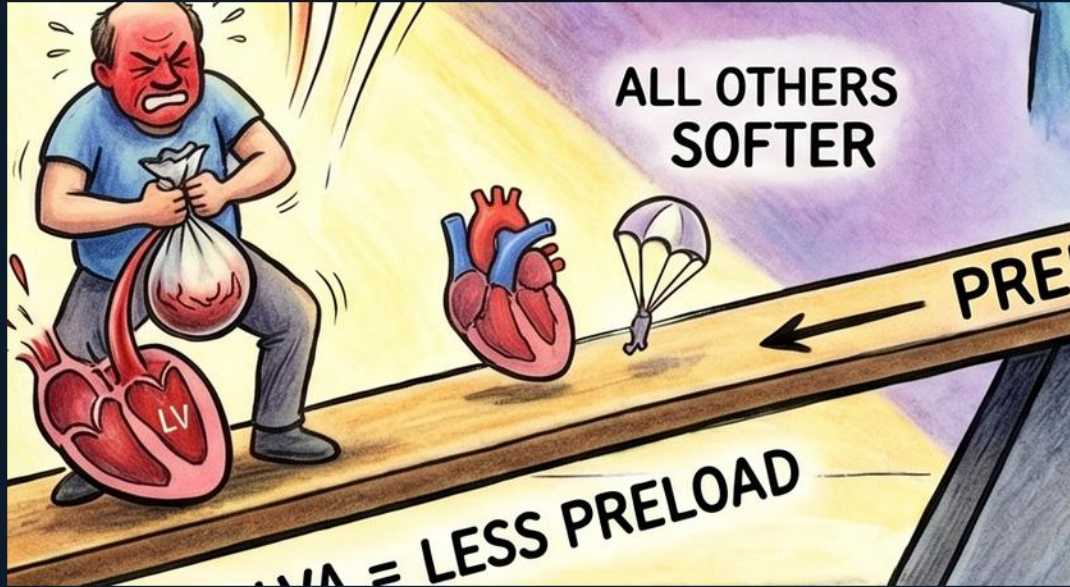
A standing figure tagged STAND LOUDER = HOCM/MVP at the fulcrum of the seesaw.

WHAT IT ENCODES

Restates that the standing position selectively augments the two exception murmurs. The seesaw fulcrum is preload: tip one way and HOCM/MVP win, tip the other and the regurgitant lesions win. Position on the seesaw equals preload state.

THE BOARD TRAP

Don't confuse standing with inspiration — standing changes preload globally; inspiration selectively boosts RIGHT-sided murmurs (Carvallo).



PRELOAD seesaw beam

WHAT YOU SEE

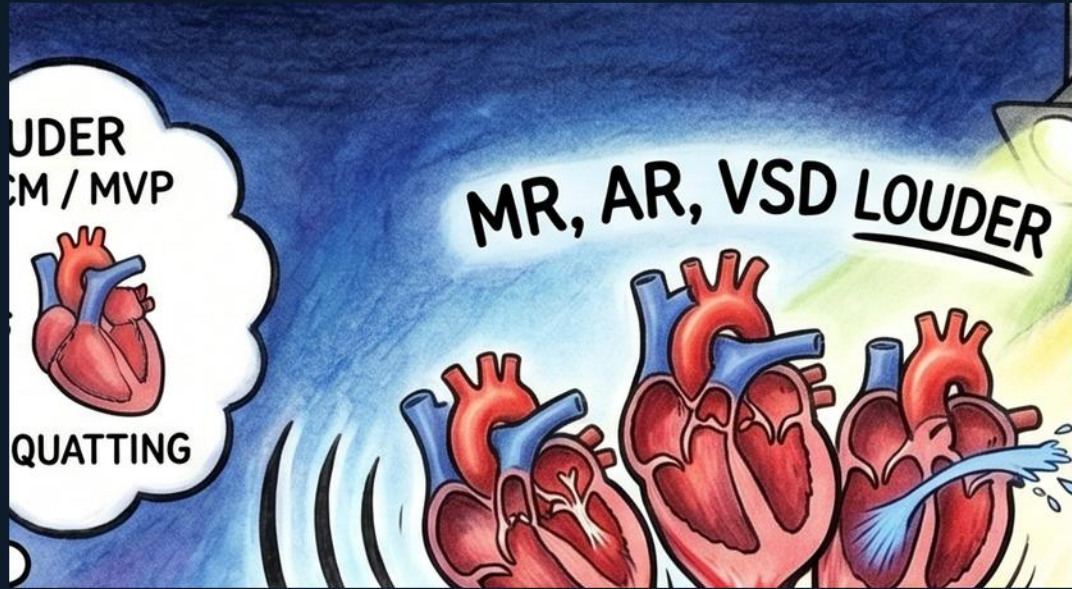
A long beam labeled PRELOAD forming the seesaw, with an arrow toward the standing side.

WHAT IT ENCODES

Preload is the single hinge variable of the whole figure: less preload (standing/Valsalva) on the left, more preload (squat/handgrip) on the right. The direction the beam tips tells you which murmurs get louder. Frame every maneuver as 'what does it do to LV filling?'

THE BOARD TRAP

Handgrip's dominant effect is raising AFTERLOAD (and preload), which is why it boosts MR/AR/VSD — don't attribute handgrip purely to preload.



MR, AR, VSD LOUDER (top-right banner)

WHAT YOU SEE

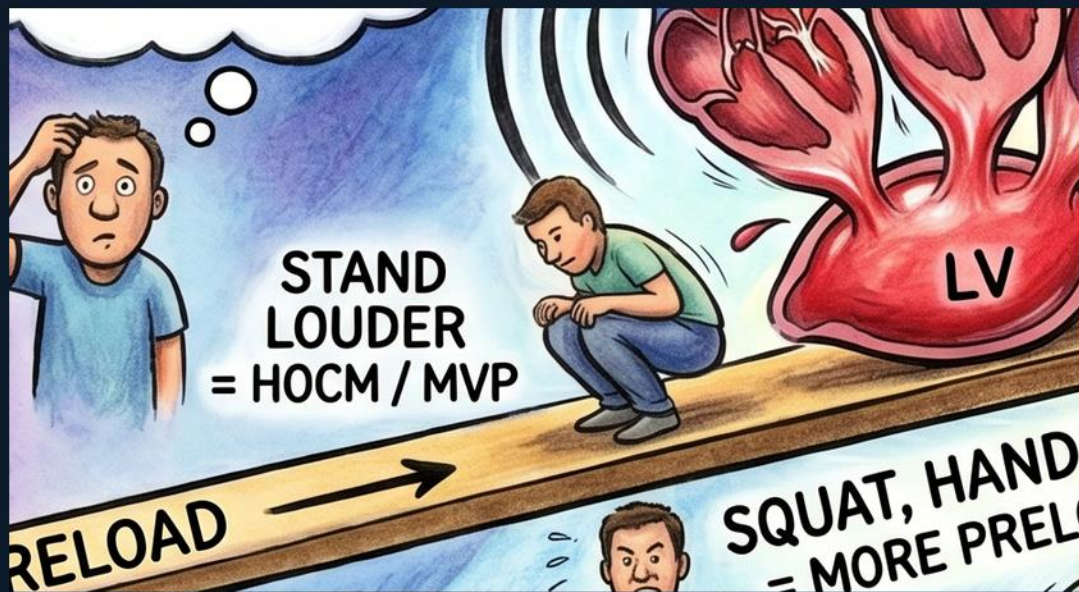
Banner reading MR, AR, VSD LOUDER over the squatting/handgrip side.

WHAT IT ENCODES

Squatting and handgrip INCREASE preload and afterload, enlarging the LV and driving more regurgitant/shunt flow, so MR, AR, and VSD murmurs grow LOUDER. Handgrip raises systemic resistance, increasing the regurgitant gradient across the aortic/mitral valve and the VSD shunt. These are the lesions that behave 'normally.'

THE BOARD TRAP

These three get louder with squatting/handgrip — if they got louder on STANDING instead, you'd be wrong: standing softens them.



Large dilated LV (red, enlarged)

WHAT YOU SEE

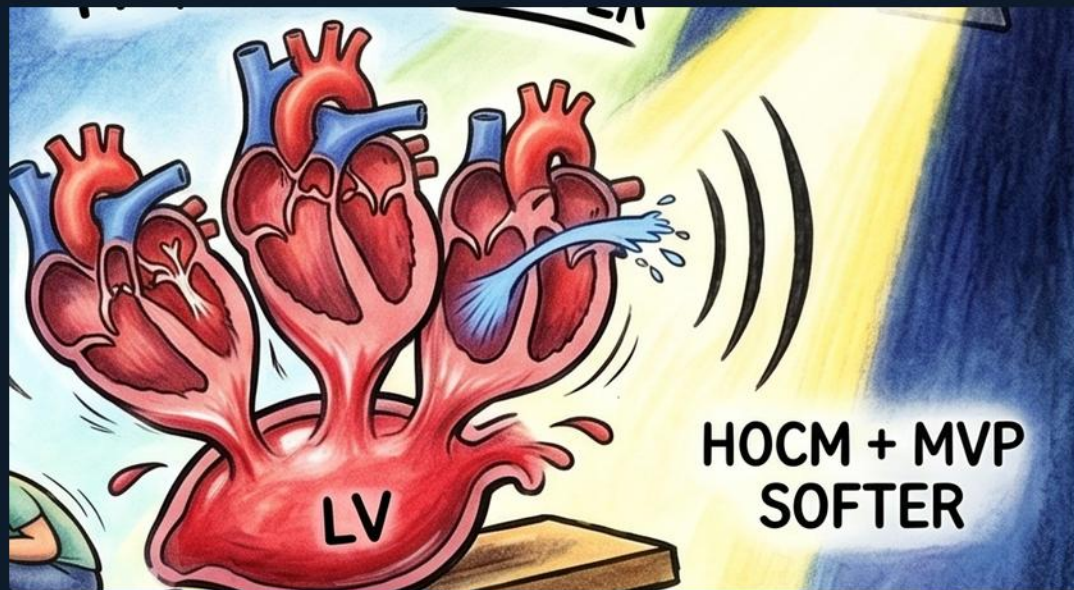
A big red heart labeled LV on the squatting side, cavity enlarged.

WHAT IT ENCODES

More preload and afterload dilate the LV. The larger cavity reduces dynamic LVOT obstruction (softening HOCM) and seats the mitral leaflets better (softening MVP), while augmenting regurgitant volume in MR/AR. Cavity size is again the deciding factor.

THE BOARD TRAP

A bigger LV makes HOCM SOFTER, not louder — a louder murmur with squatting argues against HOCM.



HOCM + MVP SOFTER (upper right)

WHAT YOU SEE

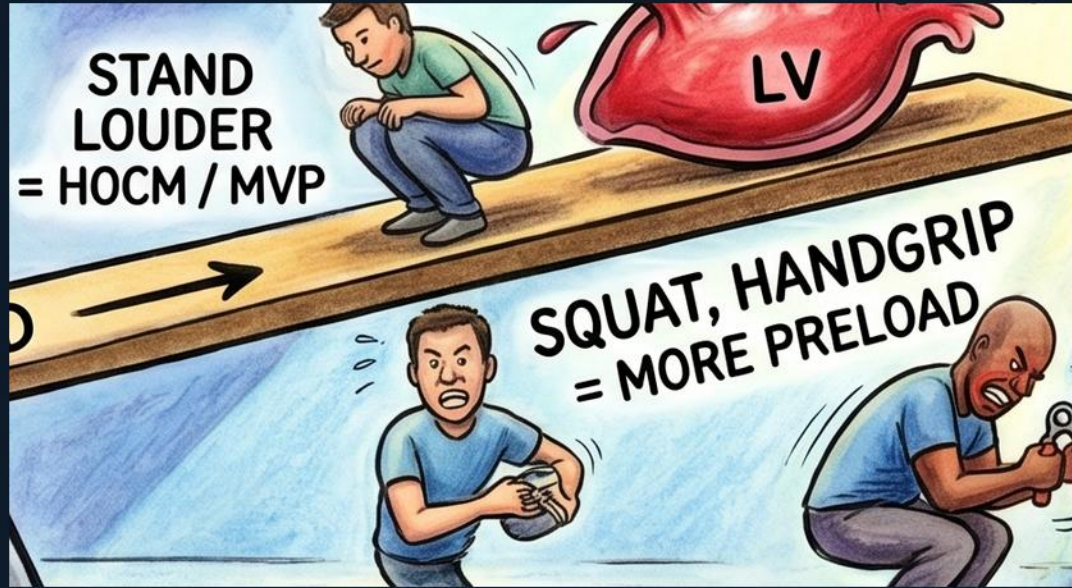
Label HOCM + MVP SOFTER beside a small heart on the squatting side.

WHAT IT ENCODES

Squatting raises preload and afterload, enlarging the LV and SOFTENING both HOCM and MVP — the mirror image of their standing behavior. In MVP the click also moves LATER (toward S2) and the murmur shortens. This confirms the exception pair tracks LV cavity size in both directions.

THE BOARD TRAP

Squatting delays the MVP click LATER and shortens the murmur — opposite of standing; reversing this is a classic error.



Squat, handgrip = more preload (figures)

WHAT YOU SEE

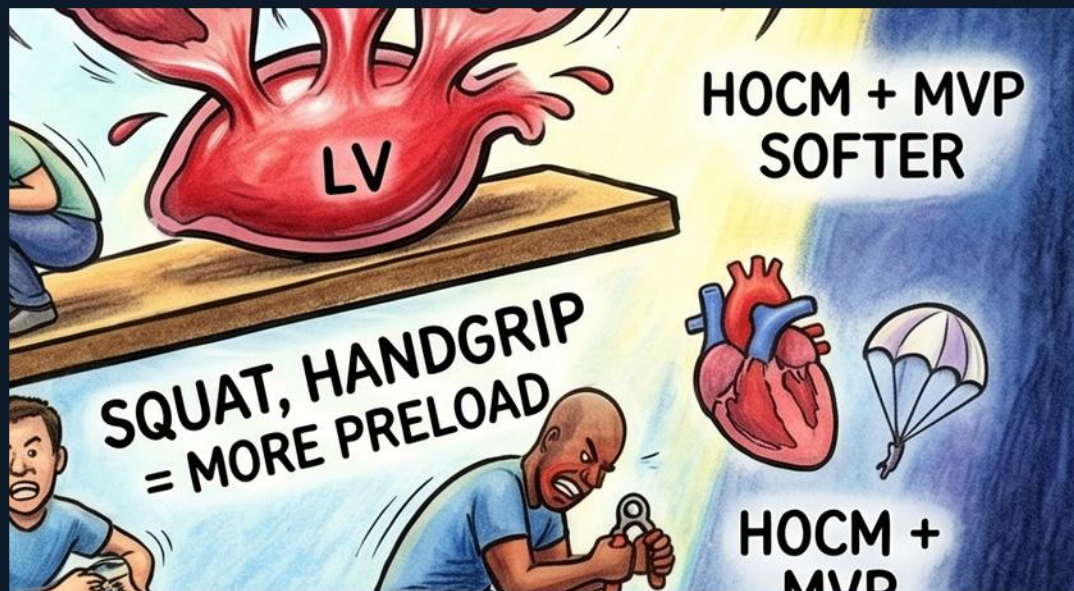
Figures squatting and gripping, labeled SQUAT, HANDGRIP = MORE PRELOAD.

WHAT IT ENCODES

Squatting increases venous return (preload) AND arterial resistance (afterload); sustained handgrip mainly raises afterload. Both enlarge the LV and increase the gradient for regurgitant lesions and VSD. They are the right-side drivers of the seesaw.

THE BOARD TRAP

Handgrip is primarily an AFTERLOAD maneuver — it loudens MR/AR/VSD and softens HOCM/AS; don't equate it with simple volume loading only.



HOCM + MVP SOFTER (lower right)

WHAT YOU SEE

A second small heart labeled HOCM + MVP SOFTER lower on the squatting side.

WHAT IT ENCODES

Reinforces that with squat/handgrip the two exception murmurs diminish. Both maneuvers enlarge the ventricle and reduce obstruction/prolapse. Keep HOCM and MVP mentally tied so you apply the exception consistently.

THE BOARD TRAP

Don't isolate HOCM from MVP — they always move together; if you make one louder you must make the other louder too.



Inspiration → right-sided heart/lung

WHAT YOU SEE

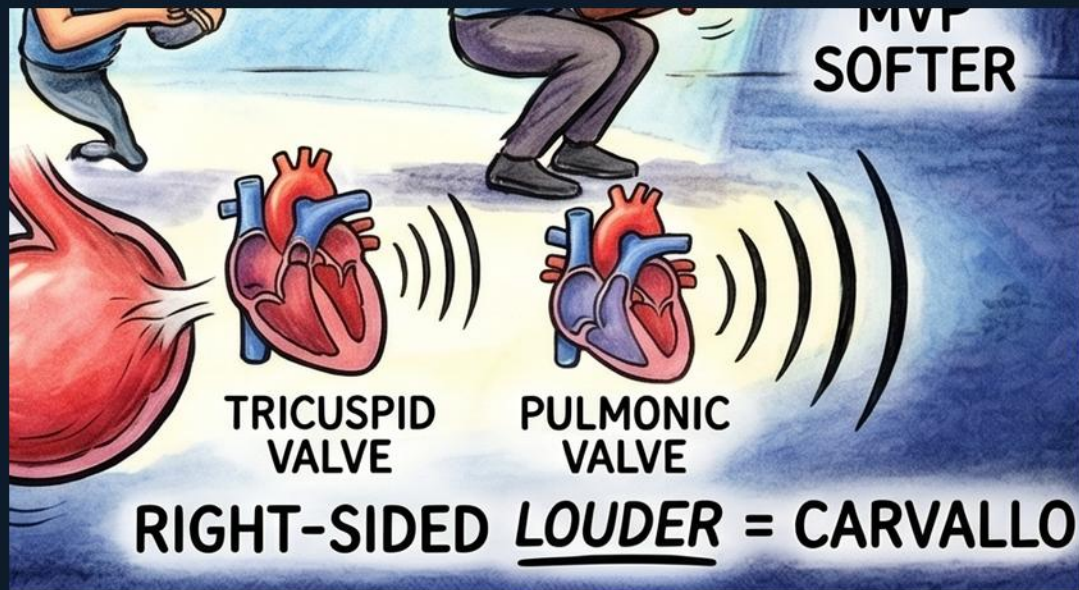
A red heart blowing into a lung under the seesaw, labeled INSPIRATION.

WHAT IT ENCODES

INSPIRATION increases venous return to the RIGHT heart, making right-sided murmurs LOUDER — this is Carvallo's sign. Tricuspid regurgitation and pulmonic murmurs intensify on inspiration; left-sided murmurs are better heard in EXPIRATION. Use the breath as the right-vs-left discriminator.

THE BOARD TRAP

Right-sided murmurs increase with INSPIRATION; left-sided with EXPIRATION — assigning a louder-on-inspiration murmur to the left heart is the trap.



Tricuspid + pulmonic valves = Carvallo

WHAT YOU SEE

Two valves at lower right labeled TRICUSPID and PULMONIC with RIGHT-SIDED LOUDER = CARVALLO.

WHAT IT ENCODES

The right-sided valves — tricuspid (regurgitation) and pulmonic — are the ones augmented by inspiration. Increased inspiratory venous return raises right-heart flow, loudening these murmurs (Carvallo's sign for tricuspid regurgitation). This is independent of the preload seesaw governing left-sided lesions.

THE BOARD TRAP

Carvallo's sign is classically described for TRICUSPID REGURGITATION — don't attribute inspiratory augmentation to a left-sided lesion like MR.

Summary — every symbol

1. HOCM + MVP LOUDER (top-left banner)

HOCM and mitral valve prolapse are the EXCEPTION pair: maneuvers that shrink the LV make them LOUDER, the opposite of every other murmur

3. Hypertrophied small-cavity heart (red, angry)

With less preload the LV cavity collapses smaller

5. ALL OTHERS SOFTER

Every murmur except HOCM and MVP gets SOFTER with standing and Valsalva because reduced preload means less flow across the valve

7. STAND LOUDER = HOCM/MVP (center figure)

Restates that the standing position selectively augments the two exception murmurs

9. MR, AR, VSD LOUDER (top-right banner)

Squatting and handgrip INCREASE preload and afterload, enlarging the LV and driving more regurgitant/shunt flow, so MR, AR, and VSD murmurs grow LOUDER

11. HOCM + MVP SOFTER (upper right)

Squatting raises preload and afterload, enlarging the LV and SOFTENING both HOCM and MVP — the mirror image of their standing behavior

13. HOCM + MVP SOFTER (lower right)

Reinforces that with squat/handgrip the two exception murmurs diminish

15. Tricuspid + pulmonic valves = Carvallo

The right-sided valves — tricuspid (regurgitation) and pulmonic — are the ones augmented by inspiration

2. Standing, Valsalva = less preload (figures)

Standing and the strain phase of Valsalva DECREASE venous return and preload, producing a smaller LV

4. Parachute = mitral valve prolapse

In MVP, a smaller LV from reduced preload lets the leaflets prolapse EARLIER — the systolic click and murmur move earlier and the murmur lengthens

6. Thought bubble: STAND LOUDER = HOCM/MVP

The mnemonic anchor: standing (less preload) = HOCM/MVP louder; squatting (more preload) = HOCM/MVP softer

8. PRELOAD seesaw beam

Preload is the single hinge variable of the whole figure: less preload (standing/Valsalva) on the left, more preload (squat/handgrip) on the right

10. Large dilated LV (red, enlarged)

More preload and afterload dilate the LV

12. Squat, handgrip = more preload (figures)

Squatting increases venous return (preload) AND arterial resistance (afterload); sustained handgrip mainly raises afterload

14. Inspiration → right-sided heart/lung

INSPIRATION increases venous return to the RIGHT heart, making right-sided murmurs LOUDER — this is Carvallo's sign